

Models and Notations

Learning objectives

By the end of this chapter, you should be able to

- identify the decision variables, objective, and feasible set;
- translate a verbal problem into an optimization problem;
- distinguish feasibility, optimality, and unboundedness;
- recognize the symmetry between minimization and maximization; and
- tell if a (feasible) set is open or closed.

Optimization asks for the best decision allowed by a collection of restrictions. Applications range from resource allocation and engineering design to portfolio selection, data fitting, and machine learning.

1.1 The anatomy of an optimization problem

Let n be a positive integer, let $S \subseteq \mathbb{R}^n$, let $f : S \rightarrow \mathbb{R}$, and write the decision vector as $\mathbf{x} := (x_1, \dots, x_n)^\top$. A general minimization (maximization) problem has the form

$$\min_{\mathbf{x} \in S} (\max) f(\mathbf{x}), \quad (1.1)$$

where the admissible decisions are the points of S .

Definition 1.1 (Basic modeling terms). In Equation (1.1), $\mathbf{x}$ is the *decision variable*, f is the *objective function*, and S is the *feasible set*. A point in S is *feasible*; a point outside S is *infeasible*.

The feasible set may be described directly or through equations and inequalities. Let m and p be nonnegative integers, let $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ for $i = 1, \dots, m$, and let $h_j : \mathbb{R}^n \rightarrow \mathbb{R}$ for $j = 1, \dots, p$; the corresponding family is empty when $m = 0$ or $p = 0$. A standard constrained nonlinear program is

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^n} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & g_i(\mathbf{x}) = 0, \quad i = 1, \dots, m, \\ & h_j(\mathbf{x}) \leq 0, \quad j = 1, \dots, p. \end{aligned} \quad (1.2)$$

The g_i 's are equality-constraint functions and the h_j 's are inequality-constraint functions. An *unconstrained* problem usually means that S is open (see Section 1.4 later) or simply $S = \mathbb{R}^n$.

Definition 1.2 (Global optimizer and optimal value). For a minimization problem (1.1),

a feasible point $\mathbf{x}^* \in S$ is a *global minimizer* if

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \text{for every } \mathbf{x} \in S.$$

The number $f(\mathbf{x}^*)$ is the *optimal value*. The set of all global minimizers is

$$\arg \min_S f := \arg \min_{\mathbf{x} \in S} f(\mathbf{x}) := \{\mathbf{x} \in S : f(\mathbf{x}) \leq f(\mathbf{y}) \text{ for every } \mathbf{y} \in S\}.$$

The set $\arg \min_S f$ may be empty, contain one point, or contain many points. When the problem has at least one global minimizer, we write $\mathbf{x}^* \in \arg \min_S f$; if uniqueness is known, write $\mathbf{x}^* = \arg \min_S f$.

For a maximization problem (1.1), its global maximizers and $\arg \max_S f$ are defined by reversing the inequality.

Important note: ultimate goal and three first questions about well-posedness

For either minimization or maximization problems, our ultimate goal is to solve for their global optimizers. However, this is impossible in general, which will be made clear in later chapters. Moreover, we shall always check the well-posedness of problems at hand, i.e., the existence of global optimizers, before solving them.

For an optimization model, three first questions about its well-posedness are: whether its feasible set is nonempty, whether its objective is bounded (in required directions), and whether its optimal value is attained. We will revisit this point in [Chapter 2](#).

1.2 Modeling examples

The main modeling work is to choose variables with clear units, express the objective in those variables, translate every restriction into a constraint, and specify minimization or maximization.

Example 1.1 (Capital and labour). Suppose K units of capital and L units of labour produce KL units of output. Capital costs 4 dollars per unit, labour costs 1 dollar per unit, and the total budget is 8000 dollars. The model is

$$\begin{aligned} & \max_{K,L} \quad KL \\ & \text{subject to} \quad 4K + L \leq 8000, \\ & \quad \quad \quad K, L \geq 0. \end{aligned}$$

Example 1.2 (A production plan). Products A and B use 2 kg and 3 kg of material per unit, and at most 200 kg are available. They require 30 and 20 minutes of machine time per unit, and the machine is available for 50 hours. Revenue per unit is 150 and 300,

while total manufacturing costs are $3x_1^2$ and $5x_2^2$. Thus

$$\begin{aligned} \max_{x_1, x_2} \quad & 150x_1 + 300x_2 - 3x_1^2 - 5x_2^2 \\ \text{subject to} \quad & 2x_1 + 3x_2 \leq 200, \\ & \frac{1}{2}x_1 + \frac{1}{3}x_2 \leq 50, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Example 1.3 (Mean–variance portfolio selection). Suppose there are n securities, where n is a positive integer. Let x_i be the fraction invested in security i , and let μ_i , σ_i , and ρ_{ij} denote expected returns, standard deviations, and correlations ($i, j = 1, \dots, n$). Define

$$\mathbf{Q} := (Q_{ij}) \in \mathbb{R}^{n \times n}, \quad Q_{ij} := \rho_{ij}\sigma_i\sigma_j, \quad \boldsymbol{\mu} := (\mu_1, \dots, \mu_n)^\top \in \mathbb{R}^n.$$

Then expected return is $\boldsymbol{\mu}^\top \mathbf{x}$ and variance is $\mathbf{x}^\top \mathbf{Q} \mathbf{x}$. For a target return R , Markowitz's model is

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{x}^\top \mathbf{Q} \mathbf{x} \\ \text{subject to} \quad & \sum_{i=1}^n x_i = 1, \\ & \boldsymbol{\mu}^\top \mathbf{x} \geq R, \\ & x_i \geq 0, \quad i = 1, \dots, n. \end{aligned}$$

Example 1.4 (Sensor localization). Let d, m, n be positive integers. Let $\mathbf{x}^1, \dots, \mathbf{x}^n \in \mathbb{R}^d$ be unknown sensor positions and $\mathbf{a}^1, \dots, \mathbf{a}^m \in \mathbb{R}^d$ be known anchors. Define the sensor–sensor measurement index set $\mathcal{N} \subseteq \{1, \dots, n\}^2$ and the anchor–sensor measurement index set $\mathcal{M} \subseteq \{1, \dots, m\} \times \{1, \dots, n\}$. For $(i, j) \in \mathcal{N}$, let d_{ij} denote the measured sensor–sensor distance; for $(i, j) \in \mathcal{M}$, let r_{ij} denote the measured anchor–sensor distance. A nonlinear least-squares model for determining the sensor positions is

$$\min_{\mathbf{x}^1, \dots, \mathbf{x}^n} \sum_{(i,j) \in \mathcal{N}} \left(\|\mathbf{x}^i - \mathbf{x}^j\|^2 - d_{ij}^2 \right)^2 + \sum_{(i,j) \in \mathcal{M}} \left(\|\mathbf{a}^i - \mathbf{x}^j\|^2 - r_{ij}^2 \right)^2,$$

An illustration is in [Figure 1.1](#). The same structure appears in molecular conformation, where the unknowns are atomic positions.

1.3 Sign symmetry and unboundedness

Proposition 1.1 (Sign symmetry). *The problems $\max_{\mathbf{x} \in S} f(\mathbf{x})$ and $\min_{\mathbf{x} \in S} -f(\mathbf{x})$ have the same optimizer set, and their optimal values have opposite signs.*

Proof. The statements $f(\mathbf{x}^*) \geq f(\mathbf{x})$ for every $\mathbf{x} \in S$ and $-f(\mathbf{x}^*) \leq -f(\mathbf{x})$ for every $\mathbf{x} \in S$ are equivalent. $\square$

Due to the symmetry above, we focus on minimization problems thereafter.

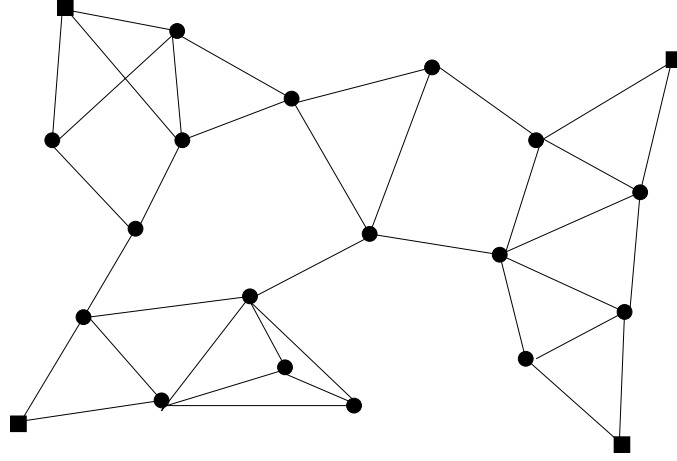


Figure 1.1: A sensor network with known anchors and measured local distances.

Definition 1.3 (Unbounded objective). The problem $\min_{\mathbf{x} \in S} f(\mathbf{x})$ is *unbounded below* if, for every $K \in \mathbb{R}$, there is $\mathbf{x} \in S$ with $f(\mathbf{x}) < K$. Equivalently, $\inf_{\mathbf{x} \in S} f(\mathbf{x}) = -\infty$. Unboundedness above is defined analogously.

Warning

An unbounded feasible set and an unbounded objective are different concepts. For example, $S = \mathbb{R}$ is unbounded, but $\min_{x \in \mathbb{R}} x^2$ has the attained minimum 0.

1.4 Open and closed sets

The geometry of the feasible set matters throughout optimization. Two topological notions will support both existence theorems and optimality conditions in later chapters.

Definition 1.4 (Open ball, open set, and closed set). For $\mathbf{x} \in \mathbb{R}^n$ and $\varepsilon > 0$, define

$$B(\mathbf{x}, \varepsilon) := \{\mathbf{y} \in \mathbb{R}^n : \|\mathbf{y} - \mathbf{x}\| < \varepsilon\}.$$

A set $S \subseteq \mathbb{R}^n$ is *open* if every $\mathbf{x} \in S$ has some $\varepsilon > 0$ for which $B(\mathbf{x}, \varepsilon) \subseteq S$. A set $S \subseteq \mathbb{R}^n$ is *closed* if its complement is open. The empty set and $\mathbb{R}^n$ are both open and closed.

Proposition 1.2 (Equivalent characterization of a closed set). A set $S \subseteq \mathbb{R}^n$ is closed if and only if, whenever $\mathbf{x}_k \in S$ and $\mathbf{x}_k \rightarrow \mathbf{x}$, the limit $\mathbf{x}$ also belongs to S .

Example 1.5 (Intervals and boxes). The interval $[a, b]$ is closed. The interval $(a, b]$ is not closed because

$$x_k := a + \frac{b-a}{k} \in (a, b], \quad x_k \rightarrow a \notin (a, b].$$

More generally, every box $\{\mathbf{x} : a_i \leq x_i \leq b_i, i = 1, \dots, n\}$ is closed.

If a set can be characterized as the equality set or sublevel set of continuous functions, we can conclude its closedness by the function continuity.

Proposition 1.3 (Continuous constraints define closed sets). *Let $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous. Then each of*

$$\{\mathbf{x} : g(\mathbf{x}) \leq 0\}, \quad \{\mathbf{x} : g(\mathbf{x}) \geq 0\}, \quad \{\mathbf{x} : g(\mathbf{x}) = 0\}$$

is closed.

Proof. Consider $S := \{\mathbf{x} : g(\mathbf{x}) \leq 0\}$. If $\mathbf{x}_k \in S$ and $\mathbf{x}_k \rightarrow \mathbf{x}$, continuity gives $g(\mathbf{x}_k) \rightarrow g(\mathbf{x})$. Since $g(\mathbf{x}_k) \leq 0$, passing to the limit gives $g(\mathbf{x}) \leq 0$, so $\mathbf{x} \in S$. The other cases are identical. $\square$

Proposition 1.4 (Finite intersections of closed sets). *Let q be a positive integer. If $S_1, \dots, S_q$ are closed subsets of $\mathbb{R}^n$, then $\bigcap_{i=1}^q S_i$ is closed.*

Proof. Let $\mathbf{x}_k$ be a sequence in the intersection with $\mathbf{x}_k \rightarrow \mathbf{x}$. For every i , closedness of S_i gives $\mathbf{x} \in S_i$. Hence $\mathbf{x}$ belongs to the intersection. $\square$

Proposition 1.4 can also be extended to the intersection of infinitely many closed sets.

Corollary 1.5 (Closed feasible set). *If the functions g_i, h_j in Equation (1.2) are continuous, then its feasible set*

$$S := \{\mathbf{x} : g_i(\mathbf{x}) = 0, \ i = 1, \dots, m; \ h_j(\mathbf{x}) \leq 0, \ j = 1, \dots, p\}$$

is closed.

Proof. Every equality set and every sublevel set is closed by Proposition 1.3; S is their finite intersection, so is also closed by Proposition 1.4. $\square$

Example 1.6 (The graph of a continuous function). The parabola $S := \{(x, x^2) : x \in \mathbb{R}\}$ can be written as

$$S = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 - x_1^2 = 0\}.$$

Since $(x_1, x_2) \mapsto x_2 - x_1^2$ is continuous, S is closed.

Example 1.7 (A nonlinear feasible set). The set

$$S := \{(x_1, x_2) : x_1^2 + x_2^2 \leq 3, \ x_1 - 2 \sin x_2 \geq 0\}$$

is closed because it is described by the continuous inequalities

$$x_1^2 + x_2^2 - 3 \leq 0, \quad -x_1 + 2 \sin x_2 \leq 0.$$

Closedness alone does not guarantee an optimizer: the feasible set may be empty or unbounded, and the objective may fail to attain its best value. The next chapter develops useful existence conditions.

Chapter summary

- An optimization model consists of decision variables, an objective, and a feasible set.
- Maximizing f is equivalent to minimizing $-f$.
- Feasibility, boundedness of the objective, and attainment are distinct and should be determined before solving the model at hand.
- Open sets contain a ball around every point; closed sets contain all sequential limits.
- Continuous equality and non-strict inequality constraints define closed sets, and finite intersections preserve closedness.

1.5 Exercises

Exercise 1.1 (Price selection). It costs a company c dollars to produce one unit. If it charges p dollars per unit, customers demand $D(p)$ units. Formulate the profit-maximization problem.

Exercise 1.2 (Widgetco). Widgetco produces widgets at plants A and B. It costs $20x^{1/2}$ dollars to produce x units at A and $40x^{1/3}$ dollars to produce x units at B. Each plant can produce at most 70 units, total production cannot exceed 120 units, and each unit sells for 10 dollars. Formulate an NLP that maximizes net profit.

Exercise 1.3 (Warehouse location). Truckco serves four customers:

Customer	1	2	3	4
Location	(5, 10)	(10, 5)	(0, 12)	(12, 0)
Annual shipments	200	150	200	300

Formulate an unconstrained NLP for the warehouse location (x, y) that minimizes total annual travel distance.

Exercise 1.4 (Closedness from sequences). Prove directly from the sequential definition that $\{(x, x^2) : x \in \mathbb{R}\}$ is closed.

Exercise 1.5 (Set operations).

- Let $S_n \subseteq \mathbb{R}^n$ be closed ($n \in \mathbb{N}$). Prove that $\cap_{n \in \mathbb{N}} S_n$ is closed.
- Let $S_1, S_2 \subseteq \mathbb{R}^n$ be closed. Prove that $S_1 \cup S_2$ is closed. How about the union of infinitely many closed sets?